

Solution Code - Engineering concepts - Algorithms_DataStructures - Task_Management_System

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: TaskManager.java

```
1 | class Task {
2 |     String taskId;
3 |     String name;
4 |     Task next;
5 |     public Task(String id, String name) { this.taskId = id; this.name = name; }
6 | }
7 |
8 | public class TaskManager {
9 |     private Task head = null;
10 |
11 |     public void add(Task t) {
12 |         t.next = head;
13 |         head = t;
14 |     }
15 |     public Task search(String id) {
16 |         Task curr = head;
17 |         while (curr != null) {
18 |             if (curr.taskId.equals(id)) return curr;
19 |             curr = curr.next;
20 |         }
21 |         return null;
22 |     }
23 |     public void delete(String id) {
24 |         if (head == null) return;
25 |         if (head.taskId.equals(id)) { head = head.next; return; }
26 |         Task curr = head;
27 |         while (curr.next != null) {
28 |             if (curr.next.taskId.equals(id)) {
29 |                 curr.next = curr.next.next;
30 |                 return;
31 |             }
32 |             curr = curr.next;
33 |         }
34 |     }
35 | }
```